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The Reinhart-Rogoff controversy as an instance of the 'emerging contrary result' phenomenon

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The article proposes a new approach to the Reinhart–Rogoff affair. In contrast to the two explanations of the controversy put forward in existing literature, I argue that *Growth in a Time of Debt* and its criticism published by Herndon, Ash and Pollin (*Does high public debt constantly stifle economic growth? A critique of Reinhart and Rogoff*) exemplify the 'emerging contrary result' phenomenon (emerging recalcitrant result, ERR). Three arguments support this hypothesis. First, the infamous spreadsheet error did not cause the findings to differ. On the contrary, the results differed mostly due to employing alternative averaging schemes. Second, the cliometric techniques employed by both research teams are justified to a similar degree. Third, the pattern in the cliometric literature focused on the 90%-threshold hypothesis suggests that the Reinhart–Rogoff controversy exemplifies an ERR phenomenon caused by submission/publication bias.

Keywords: Reinhart–Rogoff affair; emerging contrary result; ERR; cliometrics; empirical evidence; austerity

1. Introduction

Controversies and contrary results form a considerable part of econometric practice. Opinions on empirically established stylised facts often differ. According to Robert Goldfarb's (1997) rough estimate, as many as one in ten empirical articles published in *American Economic Review* instantiates the 'emerging contrary result' phenomenon. Among other fields of empirical macroeconomics, controversies appeared in studies on the effects of raising the minimum wage, capital gains taxation, unemployment spells, labour supply elasticities (Goldfarb, 1997) and empirical testing of the expansionary fiscal contraction hypothesis (Maziarz, in press).

There are several explanations of the phenomenon named by Robert Goldfarb as 'emerging contrary/recalcitrant result' (ERR). Publication bias, the term coined by Sterling (1959), is usually defined as preference expressed by journal editors for statistically significant findings. Goldfarb (1995) reformulated it in the following way: 'once an empirical result is well established, it becomes harder to publish "mere" confirmations. Researchers looking for ways to get published will search harder for refutations' (p. 231) and called it *researcher search strategy submission/publication bias*. Moreover, the publication bias exists also at the journal level. Editors are interested in articles refuting previously established stylized facts. In other words, contrary

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results emerge because of the bias for novel results present at both the demand and the supply side of econometric modelling, which are interconnected. Editors' preference for novel results makes econometricians likely to mine data in order to find methods that lead to obtaining new results from the same data-set. Similarly, econometricians' interest in novel results shapes publishing decisions.

If a controversy in empirical macroeconomics emerged on these grounds, in accordance with Goldfarb's (1997) analysis, ERR bias 'produces the following pattern: a statistically significant result is first identified, then later evidence questions the existence of the positive result' (p. 232). However, econometric results are also differentiated and contrary because of evolving methods, new data being gathered and, possibly, errors committed by economists. Therefore, not every example of contrary empirical results in macroeconomics constitutes the ERR phenomenon caused by publication bias present at the demand and supply sides of econometric modelling. There are three features of the emerging contrary result phenomenon caused by the above-defined submission/publication bias: (1) various but similarly justified methods; (2) the same or similar data sets and (3) the typical pattern in literatures.

In this article, I argue that the Reinhart–Rogoff affair instantiates the ERR phenomenon caused by submission/publication bias. Contrary to popular opinions held by both academic economists and commentators who publish in popular press,[1] I intend to demonstrate that the influence of the spreadsheet error committed by Reinhart and Rogoff (2010a)[2] is minor, as well as that the other two flaws indicated by Herndon, Ash, and Pollin (2014)[3] should be called 'methodological decisions' rather than 'mistakes' as both cliometric techniques are justified to a similar degree.

The line of argument is as follows. In Section 2, I briefly review the three explanations of the 90%-threshold hypothesis controversy present in the literature. First, some commentators either emphasize (and overestimate) the influence of the spreadsheet error on the general results or assume the averaging scheme applied by RR to be inappropriate. Second, other critics suspect that RR undertook methodological choices aimed at getting a desired result (desired by the institutions hiring them, to put it bluntly). Third, Maziarz (in press) recently argued that 'it's all in the eye of beholder', i.e. that the two approaches are justified to the same degree. In line with this approach, RR and HAP arrived at contrary results because of divergent methodological presuppositions. In Section 3, I examine the methodology underlying the Reinhart–Rogoff controversy. In Section 4, I discuss how the contradictory results were influenced by the spreadsheet error, the choice of averaging scheme and data exclusion. In Section 5, I argue that the alternative methodologies are justified to a similar degree. In Section 6, I discuss the literature on the 90%-threshold hypothesis in order to demonstrate that its pattern is typical for the emerging contrary result phenomenon created by submission/publication bias.

2. The problem as understood today

Today, there are three explanations of the Reinhart–Rogoff controversy. The first and most popular understanding of the Reinhart–Rogoff affair is based on the assumption that cliometric methodology is constant and well defined. In this case, there is only one averaging scheme that is appropriate and the one applied by HAP is believed to be it. Critics who adhere to this view mistakenly assume that RR committed the three flaws that led them to obtain the false result (e.g. Dafoe, 2014; Krugman, 2013; Reiss, 2014; HAP). Moreover, several economists focus only on the spreadsheet error. They overestimate its influence on the overall result and suspect the miscoded Excel formula of causing the contrary conclusions (e.g. Clemens, 2015; Grimson, 2014; Muslu, 2015; Stevenson & Wolfers, 2013).

Other commentators accuse RR of committing the three errors indicated by HAP *on purpose*. In line with the perspective of the economics of science, researchers face institutional limitations and incentives that motivate them to conduct analysis aimed at getting results desired from the viewpoint of political engagements of institutions employing them. In other words, the International Monetary Fund, the National Bureau of Economic Research or public universities are accused of being politically engaged and rewarding those researchers who produce evidence in favour of such conclusions supported by or useful to them. For instance, the IMF is known to hold a liberal point of view and support austerity. On the other hand, economists affiliated with public universities, due to their sources of funding, will likely support the welfare state. For example, Okalow (2013) and Wray (2013) explicitly accused RR of fudging and flubbing their research in order to produce evidence in favour of austerity. In line with this approach, RR did not commit the spreadsheet error but purposefully excluded several countries from the data-set and chose the weighted averaging scheme against which HAP argued, in order to produce evidence in favour of austerity. Such accusations of purposeful misconduct (as opposed to simple errors) were most often made by economists in the popular press. Such accusations are virtually impossible to prove and the articles in question were unlikely to have gone through a review process.

In contrast to the two previous approaches, Maziarz (in press) recently argued that the contradictory findings that RR and HAP arrived at are the outcome of different methodological commitments. Like the point of view put forth below, this explanation does not commit the fallacy that the different cliometric methods employed by RR and HAP are justified to different degrees. As shown below, arguments in favour of weighted or unweighted averaging schemes and those that focus on the exclusion of New Zealand GDP data are, to use the Keynesian saying, similarly weighted. However, Maziarz (in press) explains the divergent approaches to methodological issues by making use of the theory of thought styles formulated by Fleck (1979). Ludwik Fleck, a microbiologist and philosopher of science, believed that methodological decisions are shaped by belonging to thought collectives defined as sharing a particular educational background and exchanging thoughts with one's colleagues and co-workers.

Since this approach is not popular, I will limit my attempt to refute it to two arguments. First, taking into consideration online access to many textbooks on economics and econometrics and a great degree of unification of the demands set up by editors of the leading journals, econometric education seems presently to be coherent to a high degree. Therefore, assuming divergent opinions without concrete evidence of such divergence seems unjustified. Second, the explanation delivered by Maziarz (in press) is set up without considering Occam's razor. Namely, the same facts (similarly weighted arguments in favour of each methodology, the limited influence of the spreadsheet error) can be explained without the serious ontological commitment of bringing up the thought collectives as defined by Fleck (1979).

3. The methodological issues

Six years ago, Carmen Reinhart and Keneth Rogoff (RR) published their famous article *Growth in a Time of Debt.* They argued that there is a 90%-threshold in the relation between public debt and economic growth. The article received immediate attention not only in academia but also among practitioners of economic policy seeking evidence to support austerity due to the time of publication. RR published their now-infamous analysis when most developed economies faced historically high public debt levels caused by the recent economic crisis and multi-billion bailout plans. Krugman (2013) called it 'surely the most influential economic analysis of recent years' since a number of most prominent politicians[4] from the United States and the European Union indicated RR as the main reason for adopting austerity policies. The consensus of opinions favouring austerity changed following criticism of RR.

In 2013, Thomas Herndon, a graduate student at the University of Massachusetts was asked to replicate RR as a seminar paper. In spite of his best efforts and repeated attempts, his results differed from those reported in the original study. Finally, Herndon asked the authors of *Growth in a Time of Debt* for the original spreadsheet in order to compare the calculations. In this way, he discovered the spreadsheet error committed by RR. Namely, they excluded five countries from the key statistics due to a misspecified Excel formula that was designed to calculate average GDP growth in a sample. However, as he kept obtaining different results, Herndon consulted with his thesis advisors, i.e. Michel Ash and Robert Pollin. Herndon, Ash and Pollin (HAP), who published the working paper *Does High Public Debt Consistently Stifle Economic Growth? A Critique of Reinhart and Rogoff* in 2013. The article points out two additional drawbacks, i.e. an *unconventional* averaging scheme and selective exclusion of post-WWII data for a few countries. The paper describing the unsuccessful replication[5] became a highly influential and often cited article. HAP contributed to the change in opinion among economists. How did RR and HAP arrive at contradictory results?

RR based their cliometric research on their own data-set developed two years earlier (Reinhart & Rogoff, 2008). It consists of over 3700 annual observations of gross central government debt, inflation and the rate of GDP growth across 44 countries stretching over 200 years. *Growth in a Time of Debt* is composed of three analyses: post-war developed countries, emerging markets (1900–2009) and long-term (up to two centuries, all 44 countries). The conclusions are similar to a certain degree (however the first sample constitutes the strongest evidence in favour of the 90% threshold hypothesis). Like the discussion in the literature,[6] I focus on the analysis based only on post-war developed countries.

This sample includes data of 20 OECD countries from 1946 to 2009. RR divided 1180 yearly observations into four baskets, taking into account the debt-to-GDP ratio: below 30% (low debt), 30 to 60% (medium debt), 60 to 90% (high debt) and above 90% (very high). Then, RR calculated average GDP growth for every basket in the following way. First, they arithmetically averaged the values of economic growth for every country. Second, they calculated the arithmetic mean of every country's average pace of economic growth (cf. Equation (1)). RR concluded that 'when gross external debt reaches 60% of GDP, annual growth declines by about 2%; for levels of external debt in excess of 90% of GDP, growth rates are roughly cut in half' (RR, p. 573).

Weighted averaging scheme applied by RR:

$$\hat{B}_i = \frac{\frac{\sum_{t=1}^{z_1} GDP_{t1}}{z_1} + \frac{\sum_{t=1}^{z_2} GDP_{t2}}{z_2} + \ldots + \frac{\sum_{t=1}^{z_n} GDP_{tn}}{z_n}}{n} \quad (1)$$

where, $\hat{B}_i$ – average GDP growth of ith basket, GDP_{tX} – GDP growth in year t of country X, z_n – number of periods nth country is included in a considered basket and n – number of countries in a considered basket.

At the peak of the popularity of *Growth in a Time of Debt,* HAP accused RR of committing three methodological flaws: the spreadsheet (coding) error, selective exclusion of available data and unconventional weighting of summary statistics. First, after receiving the original spreadsheet used by RR to conduct calculations, HAP discovered that one of the formulas was misspecified. Namely, RR miscoded the excel formula that was designed to calculate average GDP growth across every country in the sample (in each basket), so that Australia, Austria, Belgium, Canada and Denmark were excluded (the error is common across the four baskets in both the post-war sample and the long-term one).

Second, HAP indicated that the data-set used by RR is incomplete, i.e. the time series of several countries start after 1946. They focused on the exclusion of four yearly observations of New Zealand's GDP growth from 1946 to 1949. The observations fall into the basket of highest debt-to-GDP ratio and omitting them biases the results in favour of the threshold hypothesis. Third, HAP argued that the averaging scheme applied by RR is 'non-standard' since each country in each basket influences the average GDP estimate to the same degree. The line of HAP's argument is as follows:

> real GDP growth in the UK averaged 2.4% per year during the 19 years that the UK appeared in the >90% public debt/GDP category. This mean GDP growth figure for the UK's 19 years in this highest public debt/GDP category then counts as one country observation in generating the mean GDP growth figure for the category for all countries. (HAP, p. 266)

Therefore, the influence of UK's average and a single country/year observation of a country that fell into the highest debt-to-GDP category only once was the same in spite of the different number of observations. Unlike RR, HAP calculated the average GDP growth for every basket applying the unweighted averaging scheme (cf. Equation (2)), so that every country/year observation influenced the estimates to the same degree.

Unweighted averaging scheme applied by HAP:

$$\hat{B}_i = \frac{\sum_{x=1}^{n}\sum_{t=1}^{z} GDP_{tx}}{n \times z} \tag{2}$$

where, $\hat{B}_i$ – average GDP growth of ith basket, GDP_{tx} – GDP growth in year t of country X, z_n – number of periods nth country is included in a considered basket and n – number of countries in a considered basket.

In response to the criticism, Reinhart and Rogoff (2013a, 2013b) apologized for the coding error and defended their methodology, indicating the reasons for their decisions and delivering a few arguments supporting their point of view. According to *Errata*, the observations of New Zealand's GDP growth in the years following World War II were excluded due to the existence of two contradictory data sources (delivered by the New Zealand Historical Statistics and Angus Maddison's Database). Similarly, Spain's data were excluded due to no reliable estimates for the period of 1959–1960, though HAP did not raise the issue.

Moreover, Reinhart and Rogoff (2013b) argued in favour of the averaging scheme originally applied, indicating that it is *perfectly natural* (p. 2) and was applied in order to avoid the situation when average GDP growth calculated for a basket is influenced mostly by one country. They exemplified their point of view with Greece's case, which experienced very high debt levels for 19 years since 1946 to 2009 and concluded that applying the unweighted averaging scheme would result in reducing the experience of post-war advanced economies to the cases of Greece and Japan. Moreover, the authors of *Growth in a Time of Debt* defended their choice of the weighted averaging scheme by putting forward the following argument: 'our approach has been followed in many other settings where one does not want to overly weight a small number of countries that may have their own peculiarities' (Reinhart & Rogoff, 2013b, p. 2).

Correcting for the flaws indicated by HAP strongly influenced the results, especially when it comes to the basket of the country/year observations with the highest levels of the debt-to-GDP ratio. On the one hand, there is still a decidedly reduced difference between the two baskets with the highest levels of the ratio. On the other hand, most commentators understand HAP's criticism as an argument against the 90%-threshold hypothesis. In fact, HAP explicitly wrote: 'contrary to RR, average GDP growth at public debt/GDP ratios over 90% is not dramatically different than when debt/GDP ratios are lower' (p. 257). The paper written by the economists from the University of Massachusetts brought about a change in opinion among academics and policy-makers. Preference for austerity was replaced by a Keynesian or left-wing point of view on anti-crisis policy.

4. The Influence of the methodological decisions

Both popular explanations of the Reinhart–Rogoff affair are inadequate because (1) the spreadsheet error influenced the overall results in a minor way and (2) cliometric methodology is changeable and differentiated. Below, I discuss the influence of each of the three flaws indicated by HAP on the results. In Section 5, I analyse arguments from the cliometric literature in favour of alternative cliometric techniques employed by RR and HAP and argue that (apart from the spreadsheet error that affected the findings to a limited degree) the different methods applied by RR and HAP are justified to a similar extent.

Before proceeding to the analysis, I need to highlight the fact that the combined impact of the three flaws indicated by HAP differs from the sum of their impacts considered separately. A few commentators acknowledged that committing the spreadsheet error totally falsified the results obtained by RR. This viewpoint is not justified because the exclusion of the five countries (Australia, Austria, Belgium, Canada and Denmark) from the summary statistics influenced the results only to a marginal degree. The difference equals 0.3% points in case of the very high debt basket and ranges from 0.1 to 0.2% points for the remaining baskets (Reinhart & Rogoff, 2013b).

RR and HAP obtained different results mostly because of the different averaging schemes they employed. A careful reading of Table 1 shows that the difference between the estimates obtained originally by RR and the ones corrected for the averaging scheme by HAP equals 1.7% points for the very high debt basket. In comparison, the other two issues raised, i.e. selective exclusion of certain years from the data-set and the spreadsheet error, caused a difference of 1.4 and 0.3% points, respectively.

In order to support their decision of excluding the New Zealand data, Reinhart and Rogoff (2013b) argued that choosing one of the two accessible databases changes the estimated value from 1.9% points (when the Maddison Database is employed) to 2.5% points (when the New Zealand Historical Statistics is used).

As demonstrated above, some of the hitherto offered explanations of the Reinhart–Rogoff affair are based on the flawed premise that the difference in results is caused only by the spreadsheet error. Exact analysis of RR and HAP shows however that the miscoded Excel formula biased the results insignificantly. The analysis of the combined

Table 1. The results corrected (separately) for each of the three drawbacks pointed out by HAP.

Debt/GDP ratio	Below 30%	30 to 60%	60 to 90%	Over 90%
HAP	4.2	3.1	3.2	2.2
Averaging scheme	4.2	3.0	3.2	1.7
Data exclusion	4.1	2.9	3.4	1.4
Spreadsheet error	4.0	3.0	3.0	0.3
RR (corrected by HAP for the transcription error[a])	4.1	2.9	3.4	0.0
RR (as originally published)	4.1	2.8	2.8	–0.1

[a]In addition to the spreadsheet error, HAP discovered that New Zealand's average real GDP growth was mistakenly copied between cells.
Source: Own calculations based on HAP, Table 3.

influence of the flaws indicated by HAP shows that when the two averaging methods are employed to calculate the mean GDP growth of the data-set containing the spreadsheet error and excluding New Zealand data, the difference is as high as 1.7% points for the very high debt basket, Table 1. In order to check robustness, HAP included the New Zealand data and calculated the means from the data containing the spreadsheet error and applied RR's averaging scheme. The combined difference caused by these two errors/methodological decisions equalled 1.4% points. In a similar manner, the change caused by correcting only the spreadsheet formula caused a difference of (only) 0.3% points in the highest debt-to-GDP ratio basket (Cf. HAP, Table 5).

In conclusion, the position held by some on the influence of the spreadsheet error on the overall result is mistaken. In contrast to the popular belief, the choice of averaging scheme had the most influence among the three flaws indicated by HAP. Other commentators attempted to explain the Reinhart–Rogoff affair by asserting that all methodological decisions taken by RR (i.e. averaging scheme, New Zealand data exclusion and the spreadsheet error) are mistaken. Below, I argue that both cliometric techniques applied by RR and HAP are justified to a similar degree.[7]

5. Justification of the alternative methodologies

Weighting the arguments in favour of each of the conflicting methods certainly extends the scope of this discussion, but it is useful to discuss the arguments that support each alternative. RR decided to use the weighting averaging scheme where the mean growth of each country in the four baskets influences the average to the same degree, without considering how many country/year observations were classified into a given basket. On the contrary, HAP decided to apply the unweighted averaging scheme and call the opposite decision *arbitrary and unsupportable* even though they presented the following argument in favour of RR's averaging scheme: 'it is possible that within-country serially correlated relationships could support an argument that not every additional country-year observation contributes a proportional amount of additional useful information' (HAP, p. 267). In fact, the debt-to-GDP ratio is serially correlated due to debt overhangs, when high levels of public debt are caused by slow economic development (Dafermos, 2015; Reinhart, Reinhart, & Rogoff, 2012).

Moreover, there is empirical evidence that supports the hypothesis of a causal chain. This hypothesis connects previous high levels of indebtedness and subsequent growth (Kumar & Woo, 2010). Considering that unusually high levels of public debt are caused by negative shocks (Reinhart et al., 2012), the country/year observations which follow one another should not determine the average value to the same degree. In addition, Pescatori, Sandri, and Simon (2014) conclude that what determines pace of economic growth are changes in indebtedness, not the ratio levels.

On the other hand, the averaging scheme supported by HAP is also justified by a number of arguments. First, it does not lead to a situation when a single country/year observation influences the average value to the same degree as a country that was classified into a basket for several years. Second, recent econometric literature (Bell, Johnston, & Jones, 2014; Egert, 2012; Égert, 2015a; Kourtellos, Stengos, & Ming-Tan, 2012) suggests that the determinants of economic growth differ among countries, i.e. the relation among public indebtedness and economic development should be differentiated. In this case, it would be better to draw cliometric hypotheses considering a greater number of cases, which can be done by applying the unweighted averaging scheme. However, most economists researching the threshold hypothesis tacitly support the country fixed effects hypothesis.

To sum up, there is a number of arguments in favour of each averaging scheme, therefore seeing RR's averaging method being clearly flawed and HAP's as being unequivocally right, which some commentators do, can justifiably be called a misinterpretation. Below I show that the same conclusion can be drawn from the analysis of argumentation in favour of including the New Zealand data. Moreover, I discuss the previously unconsidered[8] question whether public debt should be defined as gross central government debt or total gross external debt since this decision also makes the results differ to a similar degree.

Similarly to the choice of the averaging schemes, HAP's accusation of unjustified data omissions is not justified itself. At the time RR was preparing the analysis, no data on Spanish GDP before 1960 was available (Reinhart & Rogoff, 2013a). There were two estimates of New Zealand GDP, however (the Angus Maddison database and the New Zealand Historical Statistics). The estimates of New Zealand GDP growth during the post War World II period were inconsistent to a very high degree. Including one of the two estimates made the average economic growth of the highest debt-to-GDP ratio basket differ as much as 0.6% points. Therefore, RR decided to completely exclude this data (which reduced the mean by 0.5% points with respect to the value obtained by taking into account the higher of the two GDP growth estimates). However, it can be argued that one of the databases was more justified and should have been included in the analysis by RR.

Among the three 'errors' pointed out by HAP only one can legitimately be called such. There certainly are no arguments in favour of excluding the first five countries in the data-set (i.e. Australia, Austria, Belgium, Canada and Denmark). However, as I showed above, the spreadsheet error's influence is the least significant of the three flaws/methodological decisions that constituted the Reinhart–Rogoff affair. In fact, choosing what type of public debt should be considered, a matter undiscussed before 2015, has more influence.

RR based their research on gross central government debt due to lack of data needed to define public debt in a different way. Since cliometric research progressed over time, a database covering total gross external debt of these 44 countries since 1960 became recently available. Egert (2015b, p. 3765) conducted a research similar to RR and checked the robustness of the results. He used general government debt instead of gross central government debt. The difference which resulted from this choice amounted up to half of a percentage point.

Indeed, the point of view supported in this section was previously expressed in the cliometric literature. Hamilton (2013), a widely known econometrician interested in time series analysis, literally wrote that the point of view held by majority of commentators that one of the methods is erroneous and the other one right is in his opinion unjustified. In a similar vein, before the controversy broke out, Reinhart and Rogoff (2010b) admitted in their essay on cliometric research that methodological decisions taken by cliometricians are not based on purely econometric considerations:

> Those who have done data work know that mapping vague concepts like “high debt” or “overvalued exchange rates” into workable definitions requires arbitrary judgments about where to draw lines; there is no other way to interpret the facts and inform the discussion. (Reinhart & Rogoff, 2010b, p. 17)

6. The pattern in the literature

As noted above, the empirical techniques applied by RR and HAP are similarly justified on methodological grounds. Additionally, the influence of the spreadsheet error on the results was minor in comparison to the other two drawbacks indicated by HAP. Therefore, the popularly held view on the affair is not adequate. HAP did not arrive at different (and correct) results because of employing better or more validated methods. One of the reasons why contrary results emerge in empirical macroeconomics is the submission/publication bias. According to Goldfarb (1997), the emerging contrary result phenomenon created by submission/publication bias produces a unique pattern in the literature. First, when new data becomes accessible or economists’ interests focus on a new pair of variables, a statistically significant association is established. Second, a number of analyses confirming the previously established relation is published. Third, researchers focus on refuting a previously established result and there appear new articles showing no relation or an opposite relation to the previously observed one. Fourth, the contrary result is confirmed by several analyses.

This pattern in literature is created by social and personal interests present at both the supply and the demand side of econometric modelling. As Goldfarb (1997, p. 231) noticed, ‘once an empirical result is well established, it becomes harder to publish “mere” confirmations’ and researchers, in order to have their research published in top-ranked journals, mine the data aiming at contrary results. Goldfarb (1997) labelled this process as the researcher search strategy submission/publication bias. Additionally, one encounters the following publication bias at the journal level:

> editors are more likely to be interested in studies that show statistically significant effects of the newly studied variable, and less likely to be interested in studies showing the variable has no effect. Once the effect is established, however, editors are likely to be interested by conflicting results (…) showing no effect, or even an effect in the opposite direction. (Goldfarb, 1997, p. 231)

A review of the literature focused on the 90% threshold hypothesis shows that the findings produce the pattern described above and support the point of view presented in this article.

In 2010, RR published the first paper that raised the issue of nonlinearity by confirming the 90% threshold in the relation between public debt/GDP ratio and economic growth. In the same year, Kumar and Woo (2010) arrived at the same conclusion by applying econometric modelling instead of the cliometric practice of calculating averaging schemes. In a similar vein, Afonso and Jalles (2014) also confirmed the result and indicated that the relation between debt/GDP ratio and economic growth is stronger in those countries that suffer from higher indebtedness. Kourtellos, Stengos, & Ming-Tan (2012), on the other hand, concluded their econometric analysis by indicating that countries with weak democratic regimes more often suffer from high debt/GDP ratio levels. Baum, Checherita-Westphal, and Rother (2012) slightly reinterpreted the data. According to their model, the relation between the debt-to-GDP ratio and economic growth is positive up to the level of 67%, insignificant from 67 to 95% and negative above that, which also roughly supports the 90%-threshold hypothesis.

On 15th April 2013, HAP[9] published their criticism which, to the best of my knowledge, is chronologically the first article where authors questioned the threshold hypothesis. This article changed the consensus of opinions among economists and cliometricians. Subsequently, several articles confirmed the main conclusion that HAP arrived at. Goulas and Zervoyianni (2013) showed that high debt/GDP ratio levels hamper per capita growth only in cases of those countries that suffer from high uncertainty (measured by differentiation of GDP growth rates). Eberhardt and Presbitero (2013) conducted a robustness check and concluded their analysis by indicating that threshold levels depend on methodological decisions such as methods of estimation and model specification. In the struggle for novel results, Lee, Park, Seo, and Shin (2014) found a statistically significant threshold at the level of 30% of the debt-to-GDP ratio. In contrast, Pescatori, Sandri, and Simon (2014) argued that not the debt level but the debt trajectory influences subsequent GDP growth. Égert (2015a) estimated two thresholds at the levels of 20% and between 55 and 130% and in his later research (Égert, 2015b) showed that the levels are not robust.

7. Concluding remarks

The Reinhart–Rogoff controversy was very prominent in terms of influencing economic policy, being commented on by numerous economists and getting the attention of journalists and a wider public it attracted. There are three main arguments in favour of explaining the Reinhart–Rogoff affair as an instance of the emerging contrary result phenomenon caused by the submission/publication bias at both journal and research levels. First, as demonstrated in Section 4, it was not the spreadsheet error committed by RR (which biased the conclusion to a marginal degree), but rather the different methods (averaging schemes, mostly) that caused the results to differ. Second, as shown in Section 5, the weight of arguments (to use the Keynesian saying) supporting the cliometric techniques employed by RR and HAP is similar. Finally, the cliometric literature focused on empirical verification of the 90%-threshold hypothesis falls into the scheme described by Goldfarb (1997). Therefore, in the light of the arguments presented above, it is justified to characterize existing explanations as inadequate.

Using the 'emerging contrary result' phenomenon to shed light on the Reinhart–Rogoff affair demonstrates the necessity of a greater dose of scepticism in basing economic policy on novel econometric evidence. The submission/publication bias, indicated by Goldfarb (1997) as one of the reasons why contrary results emerge, causes a shift in economists' opinion concerning results of empirical macroeconomics. Therefore, newest empirical findings should not constitute evidence for economic policy. The Reinhart–Rogoff controversy taught us that economic policy-makers should base the interventions they conduct on well-corroborated theories and possibly mechanistic evidence (as Grüne-Yanoff (2015) argued) instead of seeking the most recent and (therefore) uncertain results. Similarly to the current practice of the medical sciences, where side effects and reactions to a new therapy are evaluated in several steps (including the gold standard of causal inference (as Cartwright (2007) called randomized controlled trials) before a new drug is accepted, economic policy-making should be based on well-developed and corroborated theories. Such practice would help in dismissing those results whose replication is impossible (due to, for instance, a spreadsheet error or an unconventional method) and suspending judgment before a lasting consensus among academic economists is established.

Today, the emerging recalcitrant result phenomenon is not well researched from the viewpoint of economic methodology. The Reinhart–Rogoff affair showed that its instances influence not only the academic discourse but also interventions carried out by governments, and, hence, the real economy. Therefore, the emerging contrary result phenomenon should be further analysed in order to identify methods of deciding whether contrary empirical findings occur due to improvement in estimation methods and new data becoming accessible or, on the contrary, the submission/publication bias. Finally, if the explanation of the Reinhart–Rogoff affair is based on the ERR phenomenon, the question about the relation between public debt and economic growth remains to be answered.

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Notes

1. Those opinions can be exemplified by the power words 'Excel Error That Changed History' from the title of an article on the controversy published by Bloomberg (Coy, 2013).
2. RR, henceforth.
3. HAP, henceforth.
4. Among others, the list includes Manuel Barroso (President of the European Commission), Olli Rehn (Vice-President of the European Commission), Angela Merkel (German Chancellor), Wolfgang Schäuble (German Finance Minister), George Osborn (British Chancellor of the Exchequer) and Paul Ryan (American Representative, the author of the report Path to Prosperity) (Botsch 2013; Smith 2007).
5. However, due to employing a different method, the article should rather be called a robustness check (Clemens, 2015).
6. HAP focused their criticism on the post-war developed countries analysis. In response, Reinhart and Rogoff (2013a, 2013b) accused them of being biased since the other two studies are

free from some of the raised issues. Similarly, a majority of econometric literature devoted to the 90%-threshold hypothesis also concentrates on the post-war developed countries.
7. I certainly do not state that the spreadsheet error may be defended on methodological grounds. Such mistakes should not happen in analyses of high academic and practical influence. However, the miscoded Excel formula biased the results insignificantly and did not undermine RR's conclusion or created the difference between RR and HAP.
8. In connection with the Reinhart–Rogoff affair.
9. Their (2014) Cambridge Journal of Economics article was originally published as a working paper (Herndon, Ash, & Pollin, 2013).

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